

Comprehensive QGIS Notes & Workflows

Part 1: GIS Foundations

Vector Data vs. Raster Data

GIS relies on two primary data models to represent the real world:

Feature	Vector Data	Raster Data
Structure	Geometric points, lines, and polygons	Grid of cells (pixels) in rows and columns
Best For	Discrete boundaries (cities, roads, properties)	Continuous data (elevation, temperature, imagery)
Common Formats	Shapefile (.shp), KML, GeoJSON	GeoTIFF (.tif), JPEG
Zooming Behavior	Remains sharp and scalable at any zoom level	Becomes pixelated (blocky) when zoomed in closely

Getting Free Spatial Data

1. Open your browser and navigate to **DIVA-GIS** (diva-gis.org).
2. Go to the "Free Spatial Data" section and select "Country Level Data".
3. Choose your target country and the subject (e.g., Administrative areas).
4. Download the resulting ZIP file and extract it to access your `.shp` files.

Part 2: Basic Operations

Adding Existing Layers

- **Vector Layer:** Go to `Layer > Add Layer > Add Vector Layer` > Browse for Shapefile > Add.
- **Raster Layer:** Go to `Layer > Add Layer > Add Raster Layer` > Browse for file (`.tif`) > Add.

- *Note:* You can manage visibility by checking/unchecking items in the **View > Panels > Layers** panel.

Creating a New Point Layer (Digitization)

1. Go to **Layer > Create Layer > New Shapefile Layer**.
2. Set Geometry type to **Point** and click OK.
3. Right-click the newly created layer in the Layers panel and select **Toggle Editing** (pencil icon).
4. Select the **Add Point Feature** tool from the top toolbar and click on the map to place your point (e.g., Jaipur).
5. Save your edits (disk icon) and toggle editing off (pencil icon).

Changing Layer Styles and Labels

- **Colors/Style:** Right-click the layer > **Properties > Symbology**. Choose your desired fill or marker colors, then click Apply.
 - **Labels:** Right-click the layer > **Properties > Labels**. Change from 'No Labels' to **Single Labels**. Select the correct column name (like **NAME_03**) from the drop-down menu and click Apply.
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Part 3: Geoprocessing & Spatial Analysis

Access these via **Vector > Geoprocessing Tools**.

Key Tools

1. **Clip:** Acts as a cookie cutter. Cuts one layer based on the boundaries of another.
2. **Buffer:** Creates a protected zone around a feature. (*Tip: Change the buffer's fill color to Transparent in Symbology to see what's underneath*).
3. **Union / Intersection / Difference:** Tools for combining or subtracting overlapping areas.
4. **Dissolve:** Merges internal boundaries that share a common attribute to create one solid outer boundary.
5. **Merge:** Combines multiple separate vector layers into one (**Vector > Data Management Tools > Merge vector layers**).

Advanced Workflow: Creating an Outer Boundary with Dissolve

1. Right-click your state layer and open the **Attribute Table**.
2. Click the **"Select features using an expression"** icon.

3. Choose **Fields and Values**, select your state column (e.g., **NAME_01**), type **=**, and double-click your target state. Click **Select Features**.
 4. Right-click the layer > **Export** > **Save Selected Features As...** and save the new shapefile.
 5. Open the Attribute Table of this new layer and click **Toggle Editing**.
 6. Open the **Field Calculator**, create a new field named "Value", and type **1** in the expression box. Click OK.
 7. Go to **Vector** > **Geoprocessing Tools** > **Dissolve**. Select the "Value" field as the dissolve field and Run to merge all inner boundaries.
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Part 4: Raster Operations

Raster Extraction (Clipping by Mask)

1. Select your target boundary (e.g., a district like Jaipur) using the Attribute Table expression method.
2. Go to **Raster** > **Extraction** > **Clip Raster by Mask Layer**.
3. Set your Raster/DEM as the Input layer and the district shapefile as the Mask layer. Run the tool.

Modifying Coordinate Reference Systems (CRS) / Reprojection

1. **Change Display Coordinates:** Go to **Project** > **Properties** > **General** and change "Display coordinates using" to Degrees Minutes.
 2. **Reproject a Raster (Warp):** Go to **Raster** > **Projections** > **Warp**.
 3. Select your layer. In Target CRS, browse and uncheck "No CRS".
 4. Search for your correct UTM zone (e.g., **WGS 84 / UTM zone 44N**) and select it. Save the output as a **.tif** file.
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Part 5: Data Conversions and Imports

- **Raster to Vector:** **Raster** > **Conversion** > **Polygonize**. Save as **.shp**.
 - **Vector to Raster:** **Raster** > **Conversion** > **Rasterize**. Select the ID field, set resolution width/height (e.g., 100), set Output Extent by selecting 'Calculate from layer'. Save as **.tif**.
 - **CSV to Shapefile:** **Layer** > **Add Layer** > **Add Delimited Text Layer**. Browse for CSV. Ensure Geometry Definition is set to the correct CRS (e.g., UTM Zone 44N). Click Add.
 - **KML to Shapefile:** **Layer** > **Add Layer** > **Add Vector Layer** > Select KML. Then right-click the loaded KML in the Layers panel > **Export** > **Save Vector Layer As** > Shapefile.
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Part 6: Basemaps and Plugins

- Go to **Plugins** > **Manage and Install Plugins** to search for new tools.
 - **QuickMapServices:** Search for this plugin. Once installed, go to **Web** > **QuickMapServices** > **OSM** > **OSM Standard** to add a street map.
 - **HCMGIS:** Search and install. Go to the HCMGIS menu > **BaseMap** > **Google Maps** to add satellite or standard Google maps.
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Part 7: Georeferencing an Image

1. Load a standard basemap (like OSM Standard) into your QGIS project.
 2. Go to **Raster** > **Georeferencer**.
 3. Click **Open Raster** and select your downloaded map image.
 4. Click the **Add Point** tool on the Georeferencer toolbar, then click a known location on the image map.
 5. Select **From Map Canvas** in the pop-up, then click the exact corresponding spot on your live QGIS map. Click OK. (Repeat for 4-5 points).
 6. Click the yellow gear icon for **Transformation Settings**.
 7. Set Transformation type to **Linear** and Resampling method to **Nearest Neighbour**.
 8. Check "Save GCP points" and "Load in QGIS when done". Click OK.
 9. Click the **Play icon** (Start Georeferencing) to apply.
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Part 8: Creating a Print Layout (Location Map)

1. Go to **Project** > **New Print Layout** and name it (e.g., "Location Map").
2. **Add Title:** Click the **Add Label** tool on the left panel, draw a box, and type your title in the Item Properties panel on the right.
3. **Add Main Map:** Click the **Add Map** tool on the left, draw a large box.
4. **Zoom & Pan Map:** Use the **"Move Item Content"** tool (fourth icon on the left) to position the map exactly how you want it inside the box.
5. **Lock the Map:** In the Item Properties panel on the right, check "Lock layers" and "Lock styles for layers". This freezes the map so you can build inset maps without messing this one up.
6. **Add Inset Map:** Return to the main QGIS window, turn on your larger regional boundary layers. Go back to the Print Layout, use Add Map again to draw a smaller box. Lock it when done.

7. **Add Map Elements:** Use the Add Legend, Add Scale Bar, and Add Arrow (North Arrow) tools from the left panel. Place them neatly and adjust their settings in the Item Properties panel. (Make sure map CRS is set to EPSG:4326 for arrows).
8. **Export:** Click the **Save Project** icon, then use **Export as Image** or **Export as PDF** from the top toolbar.

QGIS Advanced Workflows & Data Management

Part 1: Adding Layers & Basemaps

How to Add Layers

- **From the Menu:** Go to **Layer > Add Layer > Add Vector Layer** (or Raster).
- **Drag and Drop:** Simply drag your file (like a **.shp** or **.tif**) directly from your computer folders into the QGIS "Layers" panel on the left sidebar.

What are XYZ Tiles?

"XYZ Tiles" is a built-in QGIS tool that connects to base maps hosted on the internet (like standard Google Maps or satellite views). * **X and Y:** Represent the geographic coordinates. * **Z:** Represents your current zoom level. Instead of loading one massive world map, the server sends small, square image "tiles" perfectly sized for your screen. They provide the background geographic context so your specific data points don't just float in white space.

How to Add an XYZ Tile Basemap: 1. Open the **Browser Panel** (usually on the top left). If it's missing, go to **View > Panels > check Browser**. 2. Scroll down to **XYZ Tiles** and expand it. 3. Double-click **OpenStreetMap** (or drag it into your map canvas) to load the default world map. 4. **To add custom maps (like Google Satellite):** Right-click **XYZ Tiles > New Connection**. Paste the specific provider URL, name it, and click OK.

Crucial Rule: Layer Order In QGIS, layers are drawn from bottom to top. If you load a basemap and your data disappears, it's because the basemap is sitting on top of your stack like a blanket. * **The Fix:** Go to the Layers panel, click and hold your OpenStreetMap layer, and drag it to the very bottom of the list.

Part 2: QuickOSM Plugin

The QuickOSM plugin allows you to download specific OpenStreetMap vector data (like roads, buildings, or amenities) directly into QGIS.

Installation and Usage

1. Go to **Plugins** > **Manage and Install Plugins**.
 2. Search for **QuickOSM** and install it.
 3. Click the green QuickOSM icon (usually in the third row of toolbars) to open the interface.
 4. **Set the Key:** Enter the broad category (e.g., **amenity** for facilities, **highway** for roads).
 5. **Set the Value (Optional):** Enter the specific type (e.g., **university** or **college**). Leave blank to get everything under the Key.
 6. **Define Extent:**
 7. **In:** Type a specific town/city name (e.g., **Kurnool**).
 8. **Canvas Extent:** Downloads data only for the area currently visible on your screen.
 9. Click **Run Query**. The vector points/polygons will load onto your canvas.
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Part 3: Vector Data Deep Dive

Vector data represents real-world geographic features using distinct geometry (points, lines, or polygons). Every vector shape is tied to an "Attribute Table" (like an Excel sheet) holding data about that shape.

Common Vector File Extensions

- **GeoPackage (.gpkg):** The modern, default QGIS format. It is a single, clean database file that can store multiple vector layers, rasters, and projects all at once.
- **Shapefile (.shp):** The older, traditional standard. A "Shapefile" is actually a mandatory cluster of files (**.shp** for geometry, **.shx** for rendering speed, **.dbf** for the attribute table, **.prj** for the coordinate system). They must all stay in the same folder.
- **KML/KMZ:** Google Earth formats, great for web mapping.
- **GeoJSON:** Text-based spatial data, heavily used in web development.

Symbology (Styling Vectors)

Symbology is how you change default dots and blocks into meaningful icons and colors. 1. Right-click your vector layer > **Properties** > **Symbology** (paintbrush icon). 2. For uniform icons, choose **Single**

Symbol. 3. Click on **Simple Marker** to change size, fill color, and outline (stroke) color. 4. **Advanced Icons:** Change the "Symbol layer type" dropdown from Simple Marker to **SVG Marker** to access folders of real pictures (like hospital crosses or graduation caps).

Part 4: Raster Data & terrain Analysis

Raster data represents the world as a continuous grid of pixels (cells). Instead of exact shapes, every single pixel holds a numerical value (e.g., color for satellite images, or height for elevation models). The most common format is **GeoTIFF (.tif)**.

Downloading Raster Data (DEMs)

A **DEM (Digital Elevation Model)** is a raster where every pixel represents the height of the terrain above sea level. 1. Install the **SRTM-Downloader** plugin via the Plugin Manager. 2. Go to **Plugins > SRTM-Downloader > SRTM downloader**. 3. Click **Set to canvas extent** to define your target area. 4. Click Download. 5. *Note:* You will need a free API key from **OpenTopography** (portal.opentopography.org). Generate the key on their site, paste it into the QGIS prompt, and hit enter.

Terrain Analysis Tools

If you have a DEM raster, you can extract powerful terrain data: * **Slope:** Calculates the steepness of the terrain. * *Go to:* **Raster > Analysis > Slope**. * *To colorize:* Layer Properties > Symbology > Render type: **Singleband pseudocolor** > Mode: **Equal Interval**. * **Aspect:** Shows the compass direction that the slope is facing (e.g., North-facing vs South-facing slopes). * *Go to:* **Raster > Analysis > Aspect**. * **Hillshade:** Creates a 3D visual effect of the terrain by simulating sunlight and shadows based on elevation. * *Go to:* **Raster > Analysis > Hillshade**. * **Contours:** Draws line vectors connecting areas of equal elevation. * *Go to:* **Raster > Extraction > Contour**. (Or just change the original DEM's Symbology Render type to **Contour**).

Part 5: Digitization & Geometry Tools

Digitization is the process of manually tracing features from a satellite map to create your own custom vector data.

How to Digitize

1. Load a high-resolution basemap (like Google Satellite via XYZ Tiles).
2. Go to **Layer > Create Layer > New Shapefile Layer**. Choose Polygon, Line, or Point.
3. Right-click the new layer and select **Toggle Editing** (pencil icon).

4. Select the **Add Polygon/Line/Point Feature** tool from the top toolbar.
5. Click around the edges of a building or along a road to trace it. Right-click to finish the shape.
6. Open the **Attribute Table**, click **Add Field**, and type in the name or data for what you just traced. Save your edits.

Simplifying Geometry

If a digitized line or boundary has too many vertices (making the file too large or jagged): * Go to **Vector > Geometry Tools > Simplify**. This automatically removes unnecessary points while keeping the core shape intact.

Part 6: Georeferencing

Georeferencing is the process of taking a static image (like a scanned paper map or a JPEG) and pinning it to real-world geographic coordinates so it aligns properly over your basemaps.

1. Go to **Layer > Georeferencer** (or **Raster > Georeferencer** in older versions).
2. Click **Add Raster** and select your scanned map image.
3. Click the **Add Point** tool. Click a distinct intersection or landmark on the scanned image.
4. Choose **From Map Canvas**, then click that exact same landmark on your live QGIS basemap. (Repeat this for 4 corners/points).
5. Open **Transformation Settings** (yellow gear). Ensure your CRS is correct, set Transformation to Linear.
6. Click the **Play** button (Start Georeferencing).
7. *To check accuracy:* Go to the newly loaded raster's Properties > Transparency, lower the opacity to 50%, and see if the roads on the image line up perfectly with the basemap underneath.